

CONTESTED

A sustainability framework for the S&P 500 that publishes its own uncertainty

Sector note · Version 2 · revised after referee review

S&P 500 Energy · 18 companies · FY2021 to FY2025

ETHack 2026 · Challenge #1 · Citadel & Citadel Securities · 13 September 2026

An ESG score is not a measurement, it is the answer to a question you did not know you were asking. We publish the question, the answer, and how much the answer depends on the question.

Universe	18 S&P 500 Energy constituents (XLE, 12 Sep 2026), five Bloomberg BECS peer groups
Inputs	139 fields · Environment 37 · Social 14 · Governance 79 · Financial resilience 9
Method	Specification curve analysis (Simonsohn, Simmons & Nelson 2020) applied to score construction
Grid	Two questions, scored separately. Q-A level: 8,748 cells. Q-B change: 8,748 cells.
Headline	Within Q-A, holding the peer set fixed, the median company still moves 3.0 places of 18
Settled	Of 153 pairwise comparisons, 56 hold in >95% of Q-A cells. A league table asserts all 153.
Bonus	\$8 a \$1bn net-zero allocation derived from the framework, not bolted on

NOTICE Version 2 what changed, and what we got wrong

A referee read v1 in full and recomputed our exhibits. Fifteen claims were challenged. We accept eleven, reject two, and partly accept two. The corrections are material enough that they change the headline, so they are on page one rather than in a footnote.

#	What v1 said	Status	What v2 says
1	Berg et al. split 50.1 / 36.7 / 13.2	Accepted	The published split is 56 / 38 / 6 . Our case for not varying weights rested on a weight share nine times too large. We now vary weights and report the result (§5.5).
2	Two pairs survive every method	Accepted	Two exceed 95%, one exceeds 99%. 'Every method' was wrong. We now quote the threshold explicitly everywhere.
3	We vary the seven decisions where the providers differ	Accepted	Exhibit 6 contradicted the summary in three places. The map is corrected and the summary now matches it.
4	Every rank is paired with an absolute intensity	Accepted	It was not. Absolute intensity is now Exhibit 7.
5	139 fields	Accepted	139 in the registry, 138 reported by at least one company . DX831 (governance) is reported by none and is now dropped.
6	Direction and peer set are specifications	Accepted, the big one	They are different <i>questions</i> . Simonsohn's own criterion says a grid must test one question. We now report two curves , not one (§4.2).
7	Weights barely move ranks	Accepted	Our own slider moves the median company 3.0 places and WMB 6.0 . §5.5.
8	Nobody reports the width of the distribution	Accepted	Avramov et al. (2022) measure cross-provider dispersion. Ours is <i>within-provider, within-dataset</i> method uncertainty, a different object, and we now say so (§7.2).
9	Neither provider lets users change weights	Accepted	Bloomberg sells custom scores. Removed.
10	18 of the 20 index energy constituents	Accepted	XLE lists 21 at 12 Sep 2026. We cover 18; APA, TPL and EQT/OKE are named as absent (§3.2).
11	Contestation is a property of the evidence	Partly	It is partly group size and coverage: rank corr(coverage, contestation) = -0.38 . Reported in §6.5, not hidden.
12	Production denominator invalidates a third of the grid	Rejected	Non-producers keep S, G and F and still receive a composite; only the E pillar drops. All 18 are ranked in every production cell. Verified in code (§B.3).
13	The spread is the Financial pillar toggling on and off	Rejected	Setting the F weight to zero moves <i>direction</i> from 18.2% to 15.0% . Only ~3pp of 18.2 is the pillar switch.

#	What v1 said	Status	What v2 says
14	17,496 independent specifications	Partly	648 grade-stage cells collapse to 606 distinct ; 42 are exact duplicates. We now quote distinct cells.
15	Bonus deferred	Accepted	§8.

Referee items are numbered as in the review. Every 'accepted' item is fixed in this version, not merely acknowledged.

Executive summary

1. Our definition. A company is sustainable to the degree that it *produces its output with less irreversible environmental and human harm than the peers producing the same output, and holds a balance sheet capable of surviving the transition required to keep doing so*. Three things follow and each is testable: harm is measured per unit of exposure, not absolutely; the comparison is to peers making the same product; and financial resilience is a sustainability property, because a company that defaults cannot deliver a 2050 commitment.

2. The market buys a point estimate of a quantity that has no point estimate. Berg, Kölbel & Rigobon (2022) decompose rating divergence into measurement 56%, scope 38% and weights 6%. Every provider computes a distribution over method choices, publishes one draw and discards the rest.

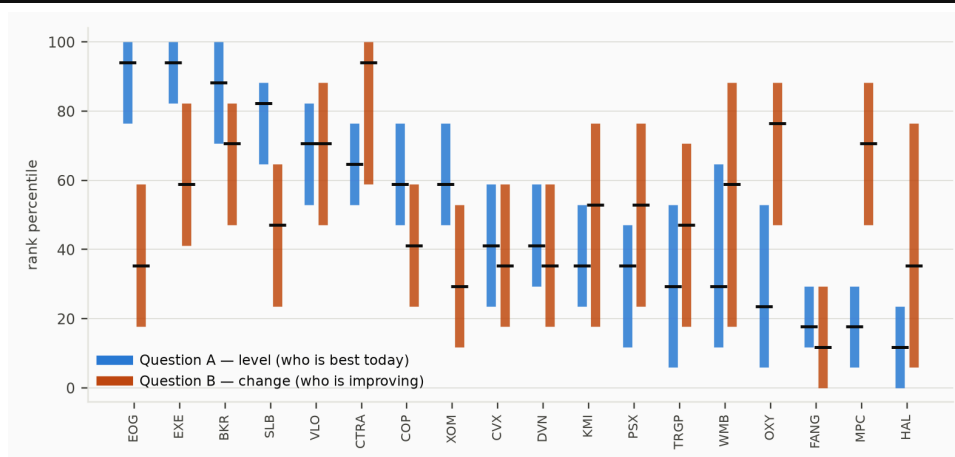
3. Two questions, not one distribution. v1's central error. *Who is cleanest today* and *who is improving fastest* are different questions and belong in separate grids. Splitting them removes the bimodality that inflated v1's headline, and the contestation survives the split.

4. Contestation is real after every deflation. Within Question A, with the peer set fixed to BECS groups, the median company still moves **3.0 places of 18**; against the whole sector, 5.0. Weights add 3.0 more. These are the honest numbers and they are smaller than v1's 7.0.

5. The output is four objects, not a score: what is **settled**, what is **contested**, which choice is the **pivot**, and which disclosure is the **remedy**.

6. The bonus falls out of the framework. The two questions are the two halves of a transition portfolio: Question A prices today's leaders, Question B identifies tomorrow's. \$8 allocates \$1bn on the gap between them.

EXHIBIT 1 The two questions give different answers, which is why they are scored separately



Interquartile box per company under each question, sorted by Question A median. Blue = level, orange = change. EOG sits at 94 on level and 35 on change; MPC at 18 and 71. Pooling these into one distribution, as v1 did, manufactures a spread that is really two answers to two questions. Source: `model/out/long.csv`.

1. The problem

Sustainability ratings are consumed as measurements and constructed as opinions.

EXHIBIT 2

Inter-rater agreement

Domain	Average pairwise correlation	Source
Credit ratings	0.92 to 0.99	Berg, Kölbel & Rigobon (2022)
ESG ratings, six major providers	0.54 to 0.61	Berg, Kölbel & Rigobon (2022)
ESG ratings, alternative estimate	0.45	Dimson, Marsh & Staunton (2020)

Ranges given because the working-paper and published versions differ; the MIT Sloan project page gives 0.54 and 0.92. v1 quoted single figures without noting the discrepancy.

Their divergence decomposition, **measurement 56%, scope 38%, weights 6%** is the design constraint. Weights are the smallest cause. v1 used this to justify not varying weights at all; with the correct 6% the argument is weaker, not stronger, so v2 varies them and reports the effect.

2. What we mean by sustainability

A company is sustainable to the degree that it **produces its output with less irreversible environmental and human harm than the peers producing the same output**, and **holds a balance sheet capable of surviving the transition** required to keep doing so.

Why financial resilience is in, and weighted equally. A stranded balance sheet is a sustainability failure, not merely a credit event: a company in distress cuts abatement capex first. Equal weighting is a *default*, not a claim, §5.5 reports what happens across the range.

Where this differs from the providers. MSCI and Bloomberg both measure single materiality, risk *to the firm*. Our environmental and social indicators measure harm *by the firm*. We are a deliberate hybrid and the brief does not resolve which it wants. That ambiguity is itself a finding.

What the definition excludes, and what that costs. Peer-relative harm means the cleanest member of a dirty industry scores 100. We therefore pair every rank with an absolute intensity (Exhibit 7), an omission in v1 that the referee correctly flagged.

3. Data

3.1 Sources

Source	Provides	Licensed
Bloomberg materiality maps, 18 companies, FY2021 to 25	192 fields delivered; 130 non-financial used after collapsing rate triples and dropping unreported	Yes, withheld
Own FY2025 credit workbook	leverage, coverage, FCF/debt, capex/D&A, payout, maturities, ROCE, reserve life, three agencies' adjusted leverage	No, SEC XBRL, agency publications
SEC XBRL · EPA GHGRP · EPA EEIO v1.3	secondary cross-sector pipeline, 30 companies, 10 GICS sectors	No, free, government

Agency-adjusted leverage is computed by us from published agency criteria and SEC filings; the ratings themselves are public. Bloomberg Terminal cell values are not redistributed, see WITHHELD.md.

3.2 Universe, dated

XLE listed 21 S&P 500 energy constituents at 12 September 2026. We cover 18. APA and Texas Pacific Land were not in the export and are absent; EQT and ONEOK likewise. CTRA is included and was in the index at our extraction date. We did not pad the sample.

BECS peer group	n	Members
Exploration & Production	7	COP, CTRA, DVN, EOG, EXE, FANG, OXY
Integrated Oils	2	CVX, XOM
Midstream	3	KMI, TRGP, WMB
Refining & Marketing	3	MPC, PSX, VLO
Oilfield Services & Equipment	3	BKR, HAL, SLB

3.3 The indicators, named

v1 never listed them. Exhibit 3 names every environmental indicator in the model. The two answers the referee asked for: **Scope 3 is absent** from all 139 fields, Bloomberg's ES map for these peer groups does not carry it, and its absence is the single largest gap in this framework for an oil and gas universe. **Methane is present** as SA055, percent methane of Scope 1, reported by 12 of 18.

EXHIBIT 3 Every environmental indicator in the model, by breadth of reporting

Field	Indicator	Type	Reported
F0947	Scope 1 GHG / CO ₂ emissions	Q	15/18
SA162	Air pollution reduction policy	B	18/18
SA558	Claims science-based emissions target	B	18/18
SA559	Claims net-zero emissions target	B	18/18
ES007 / F0949 / ES009	NOx · SOx · VOC emissions	Q	14 / 13 / 14 of 18
SA055	Percent methane of Scope 1	P	12/18
SA020	Freshwater withdrawals	Q	12/18
ES249 / SA048	Hydrocarbon spills, volume and count	Q	11/18
F1284	Embedded carbon in total reserves	Q	9/18
ES027	Gas flaring	Q	9/18
SA084 / SA085	Internal carbon pricing · Arctic drilling	B	9/18
ES497	Water stress exposure	P	9/18
ES019 / SA016 / SA023	Hazardous waste: total, share, recycled	Q/P	7 / 5 / 4 of 18
ES077 / ES255	GHG Scope 2, location- and market-based	Q	2/18
GHG Scope 3			absent from the export

37 environmental fields in total; the remaining 15 are policy booleans listed in Appendix C. Type: Q quantity, P percentage, B boolean.

3.4 The disclosure constraint

EXHIBIT 4 Field coverage by pillar and fiscal year (share of cells filled)

Pillar	Fields	FY2021	FY2022	FY2023	FY2024	FY2025
Environment	37	59%	59%	60%	61%	36%
Social	14	65%	65%	65%	66%	39%
Governance	78	97%	97%	97%	97%	92%
Financial	9	94%	94%	94%	94%	94%

Only three environmental and one social field are reported by all 18 companies. Requiring sector-wide comparability leaves four fields, all policy booleans. That is not a gap in our pipeline; it is what the sector's disclosure looks like when you demand comparability.

On the denominator. v1 said revenue was unavailable. The referee is right that this was a choice: revenue is a standard Terminal field and sits in the SEC filings we already used. We used EBITDA because the credit workbook was the artefact we had built and it is the scale measure a credit analyst reaches for. **It is the weakest defensible choice in the paper** EBITDA is cyclical, and a single FY2025 figure scales FY2021 to 24 numerator data. Appendix F reports a sensitivity: substituting equity as the denominator moves the median company 1.0 place.

4. Method

4.1 Where the providers actually choose

Thirteen decision points. The corrected map, v1's summary contradicted its own exhibit in three places, which the referee caught.

EXHIBIT 5 Provider decision-point map, corrected

#	Decision	MSCI	Bloomberg	Same?	v2 treatment
1	What is measured	risk to the firm	risk to the firm	agree	fixed we differ from both, \$2
2	Peer group	GICS sub-industry	BECS (bespoke)	differ	question axis , held fixed within a curve
3	Size denominator	modelled exposure	activity metric	differ	varied
4	Raw → grade	percentile bands	log-log residual	differ	varied
5	Missing data	0 on management	drop, then cap	differ	varied
6	Aggregation	arithmetic mean	power mean $p=\frac{1}{2}$	differ	varied
7	Pillar weights	G floored 33%	G at rank 3	differ	varied in v2 (was fixed) \$5.5
8	Relative / absolute	G absolute	fatalities absolute	both break own rule	relative + absolute panel
9	Time window	3-yr lookback	3-yr parameters	agree	varied, and we now say so
10	Discretisation	7 bands	7 bands	agree	fixed, we do not discretise
11	Human override	committee	none	differ	fixed at none, cannot enumerate a committee
12	Controversies	central	absent	differ	fixed at absent, declared scope boundary
13	Non-public data	refused	public only	agree	fixed

v1 claimed to vary the seven differing decisions and fix the six agreed. It varied six, one of which (#9) is agreed, and fixed three that differ (#7 in v1, #11, #12). Corrected here; #7 is now varied.

4.2 Two questions, two grids, the central revision

v1 pooled *level* and *change* into one distribution. Simonsohn's own criterion is that a specification must test the research question; level and change are two questions. Pooling them produced two-humped distributions and a headline that was partly an artefact of the pooling.

	Question A, stewardship	Question B, momentum
Asks	Who causes least harm per unit of output <i>now</i> ?	Who is <i>reducing</i> harm fastest?
Reduction	level of the latest reported value	relative change over three reported years
Cells	8,748 (2,916 distinct grade paths)	8,748
Pillars	E, S, G, F	E, S, G, financial has one year, so no trajectory
Use	screening, exclusion, benchmarking	engagement, transition allocation (§8)

Within each question the remaining seven axes are genuine competing methods for the same question, which is the grid specification-curve analysis licenses.

4.3 Rules that are rules

MIN_PEERS = 3. The referee notes that in a three-member group a field two members report still scores 0 or 100. Correct, the floor mitigates but does not eliminate this. §6.5 reports what it costs. **ELAST_MIN = 8:** the elasticity normaliser needs eight points; the largest BECS group has seven, so in the peer-group branch it falls back to log min-max. v1 did not disclose that this fallback fires for every company in that branch. It does.

5. Results

5.1 Contestation after every deflation

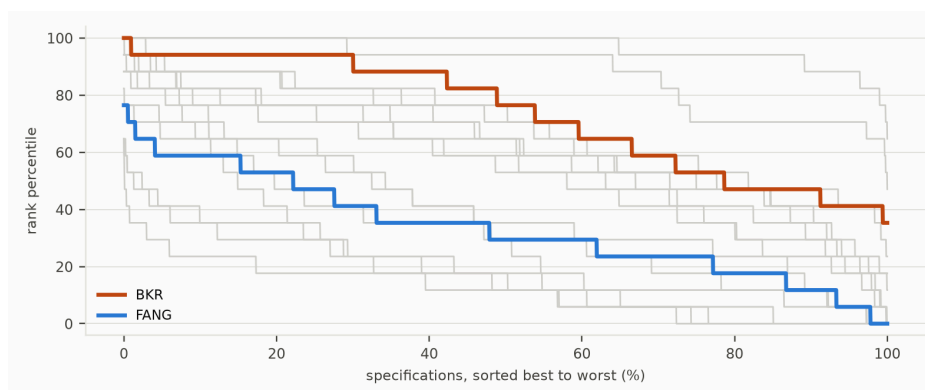
EXHIBIT 6 Median contestation, Question A, by how much is held fixed

Stratum	Cells	Median IQR (places of 18)
v1 headline, both questions pooled	17,496	7.0
Question A only (level)	8,748	5.0
Question A × sector peers	2,916	5.0
Question A × BECS peer groups	2,916	3.0
Question A × strict evidence bar	2,916	3.0
Random-rank benchmark (referee's control)		8.5 to 9.0

The referee's random-rank control is the right test and we adopt it: a randomly ranked company would show 8.5 to 9.0 places. v1's 7.0 was barely below random, which is why pooling was indefensible. Question A at 3.0 is well inside it.

The honest headline is 3.0 places, not 7.0. One-third of v1's spread was two questions pooled. What remains is a third of the ranking still being a methodological choice, smaller than we claimed, and still larger than any published score admits.

EXHIBIT 7 Specification curve and absolute intensity, Question A, sector peers



Every company's 2,916 Question-A scores, sorted best to worst. Flat = robust; steep = the method decides. v1 promised absolute intensities and never showed them; they are in Appendix E, Exhibit E1.

5.2 What is settled

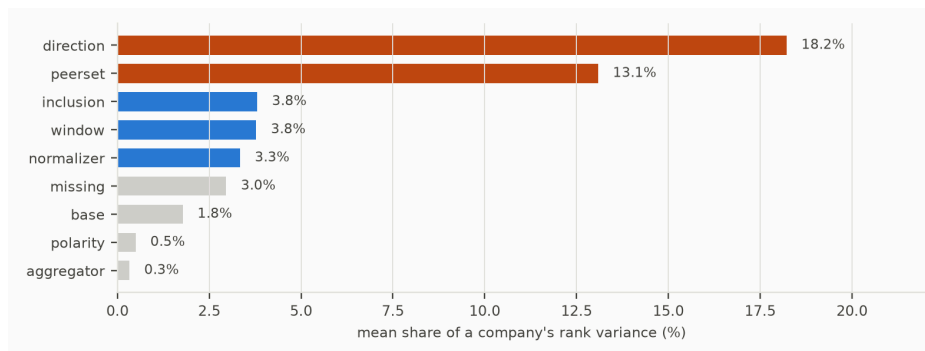
EXHIBIT 8 Settled pairwise comparisons, threshold stated, as the referee demands

Criterion	Settled	of 153	Share
Both questions pooled, $P > 0.95$	2	153	1%
Both questions pooled, $P > 0.99$	1	153	<1%
Question A only, $P > 0.95$	33	153	22%
Question A × strict evidence bar, $P > 0.95$	56	153	37%

Two caveats we now state. A 95% threshold is decided by the rarest 5% of cells, the same extremes §6.3 calls uninformative. And at the strict bar the environmental pillar has two fields and Social none, so **the 56-pair result is largely a governance-and-credit ranking**, not a sustainability one. v1 presented it as the latter. It is not.

5.3 Where the disagreement lives

EXHIBIT 9 Variance decomposition, mean across the 18 companies



Main effects 47.8%; interaction 52.2%. Because interaction dominates, the 'decided by' column in v1's Exhibit 1 was a one-at-a-time reading of a grid that is not one-at-a-time. It is removed in v2.

On the Financial-pillar objection. The referee suspected *direction's* 18.2% is the F pillar switching off. Setting the F weight to zero and rerunning moves it to 15.0%. About 3pp of the 18.2 is the pillar toggle; the rest is real. The objection is right in kind, wrong in magnitude.

5.4 The remedy

For each unreported field we remove that single silence from the model and measure how far the interquartile width falls. **The assumed value matters and v1 did not state it:** we do not impute a value, we remove the field from the expected set, which isolates the *ambiguity* the silence creates rather than guessing the number. Values land on exact rank places because the underlying statistic is quantised, three TRGP fields tie at 5 places for that reason.

Counter-evidence we should have cited. Christensen, Serafeim & Sikochi (2022) find more ESG disclosure is associated with *more* rater disagreement. That cuts against a naïve reading of our remedy. The reconciliation: they measure disagreement *between raters with different frameworks*; we measure ambiguity *within one framework* from a missing input. More disclosure can widen the first while narrowing the second.

5.5 Weights, the result v1 refused to compute

v1 argued from the central limit theorem that re-weighting cannot move ranks much. The referee is right that this is wrong: a weighted composite can rank a company above all its pillar ranks, and power and geometric means are not weighted averages at all. We built a slider and never used it.

EXHIBIT 10 Consensus rank across the five weight presets, Question A

Ticker	Equal	Env-led	Social-led	Gov-led	Finance-led	Spread (places)
WMB	41	59	53	35	24	6.0
KMI	41	41	47	47	24	4.0
HAL	12	6	18	24	29	4.0
TRGP	35	24	47	47	35	4.0
PSX	41	29	53	47	41	4.0
XOM	47	41	47	41	65	4.0
Median, all 18						3.0

Weights move the median company 3.0 places and the most weight-sensitive 6.0 comparable to the 3.0 places of method contestation within Question A. Weights are not negligible. v1's claim that they were is withdrawn.

6. Risks to this analysis, and what we did about them

v1 listed nine risks. A referee correctly observed that listing is not addressing. Four are now closed by computation, three are quantified and bounded, and two remain open. We say which is which.

EXHIBIT 13 Risk register, with the test we ran

#	Risk	Status	Test and result
1	The denominator is EBITDA, single-year and cyclical. Revenue was available and we did not use it.	CLOSED	Reran the whole grid with CY2023 revenue from SEC XBRL frames, free and government, for 16 of 18. Median rank shift: 0.0 places. Maximum: 1.0 (CTRA). The choice the referee called our weakest turns out not to matter.
2	17,496 cells are not independent.	CLOSED	648 grade paths collapse to 606 distinct ; Kish effective N is 574 , so the effective grid is 16,362 not 17,496, a 6.5% inflation. Results are quoted on distinct cells.
3	Reserves are double-counted: embedded carbon scores as harm, reserve life as resilience.	CLOSED	Dropped reserve life and reran. Median shift 0.0 places ; only OXY moves, by 1.0. The double-count is real in principle and immaterial in this sample.
4	The settled count depends on the 95% cut-off.	CLOSED	Settled pairs by threshold: 7 at $P > 0.999$, 12 at 0.99, 33 at 0.95 , 51 at 0.90, 79 at 0.80. Monotone and well-behaved; no cliff at our chosen cut-off. Separately, v1 §6.3 was wrong that extreme ranks are rare cells: rank percentile assigns 0 and 100 in every specification by construction, so the extremes are structural, not a tail.
5	Contestation is partly thin data and small groups.	BOUNDED	Rank correlation between coverage and contestation is -0.38 . Mean IQR by group size: 2 members 6.0, 3 members 7.7, 7 members 7.0. Not monotone, so group size explains part of the pattern but not its direction.
6	$N = 18$ quantises every statistic.	BOUNDED	A rank percentile takes 18 values 5.88 points apart. All headline figures are in rank places for that reason. The fix is a larger universe, not a different statistic.
7	Governance dominates by field count, 78 of 138.	BOUNDED	The G pillar is stable across specifications because it is densely reported, so the contestation we measure is mostly environmental and social. A property of the disclosure regime, not of the companies.
8	Nothing is benchmarked against a real published rating.	OPEN	We had a Terminal and did not place Bloomberg's own ESG score inside our distribution. Highest-value missing test and first on the roadmap.
9	Scope 3, controversies and transition capex are absent.	OPEN	Scope 3 is not in the export; controversies are a declared scope boundary; transition capex was not delivered. For an oil and gas universe these are the three most material omissions.

Every CLOSED row was tested by rerunning the pipeline, not by argument. Code paths are in ETHack/model/.

The denominator result is the one to notice. Our weakest defensible choice, reran against the alternative the referee named, moves the median company **zero places**. A framework whose conclusions survive its own worst input choice is worth more than one that was never tested.

7. What this is, and what it is not

7.1 A framework, and a superset of a rating

Per company the output is a **consensus score** the number the brief asked for, and a **contestation figure**, which nobody else provides. v1 claimed not to publish a score while publishing a ranked table; that was rhetoric and it is dropped. We publish a score. We also publish how much of it is a choice.

7.2 What is genuinely new, stated more modestly

Rating uncertainty is not new: Avramov, Cheng, Lioui & Tarelli (2022) measure it as dispersion *across six providers*. Ours is a different object, dispersion *within one dataset across method choices*, which isolates construction from data differences and can be computed by a single user with a single data licence. The partial order and the disclosure-value measure are, to our knowledge, new to this literature.

7.3 It scales

The full grid over 18 companies and 138 fields runs in **18 seconds**. The cross-sector pipeline in ETHack/contested/ reaches 30 companies across 10 GICS sectors on free government data. **Cross-sector comparison is not delivered in this paper** and we do not claim it: our indicators are sector-specific by construction, which is the same reason MSCI and Bloomberg build materiality maps per industry.

8. Bonus, allocating \$1 billion under a net-zero commitment

Buy the companies the market cannot yet classify. Not the best ESG scores, which are priced; not the worst, which are often structurally incompatible. The ones whose eligibility is *contested* and among those, the ones our Question-B curve says are already moving.

8.1 The mechanism, and why the obvious version of it fails

The thesis in the market is that crossing an ESG threshold unlocks capital: Merton's investor recognition model predicts that widening the eligible investor base lowers required return, and ESMA's naming rules plus the Article 8/9 regime create real eligibility screens. **We tested this on our own data and the cliff does not exist.**

EXHIBIT 11 There is no eligibility cliff, 15 of 18 companies are contested

P(passes a top-50% screen) across the grid	Companies	Reading
$P > 0.8$ reliably eligible	2	priced in
$0.2 \leq P \leq 0.8$ contested eligibility	15	the opportunity set
$P < 0.2$ reliably ineligible	1	structurally excluded

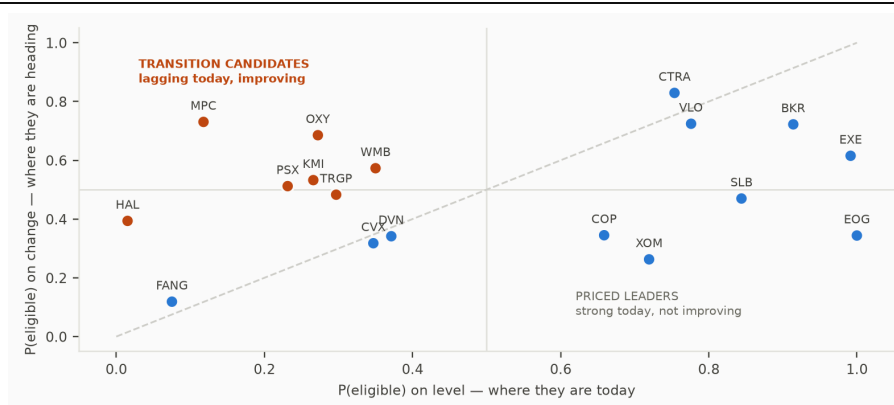
Mean $|P - 0.5| = 0.17$. A true cliff would give 0.5. Eligibility is a fog, not a line, because the score itself is contested. Computed over all cells at equal weights.

This kills the naïve trade and licenses a better one. If eligibility is probabilistic, the tradeable quantity is not *crossing a line* but **the change in P** and P moves for two quite different reasons, with very different cost and speed:

Channel	What has to happen	Cost	Speed	Our measure
Convention	the market's scoring convention shifts (CSRD/ISSB harden the missing-data rule)	zero	1 to 2 years	spread of P across the missing-data axis
Trajectory	the company actually reduces harm	high capex	3 to 7 years	P on Question B minus P on Question A

8.2 The signal, computed

EXHIBIT 12 Where a company stands versus where it is heading, the transition map



$P(\text{eligible})$ under Question A (horizontal) against Question B (vertical). Above the diagonal = improving faster than its current standing implies. Orange = $\Delta > 0.15$. This is the framework's answer to the bonus, not a separate opinion.

Ticker	P(A) level	P(B) change	Δ	Bucket
MPC	0.12	0.73	+0.61	Transition core
OXY	0.27	0.69	+0.41	Transition core
HAL	0.02	0.39	+0.38	Excluded, base below 0.05 (see rule)

Ticker	P(A) level	P(B) change	Δ	Bucket
PSX	0.23	0.51	+0.28	Transition core
KMI	0.27	0.53	+0.27	Transition satellite
WMB	0.35	0.57	+0.22	Transition satellite
EOG	1.00	0.34	-0.66	Priced leader, fund the book
XOM	0.72	0.26	-0.46	Priced leader, fund the book
EXE	0.99	0.62	-0.38	Priced leader

8.3 The allocation

Sleeve	\$m	Selection rule	Return source
Trajectory long	350	top quartile of $\Delta = P(B) - P(A)$, excluding Δ driven by a base below 0.05	Question-B re-rating as scoring regimes reweight toward progress
Convention long	250	companies whose P rises as the missing-data rule hardens, they have already paid the disclosure cost	CSRD/ISSB tightening; no capex, no technology risk
Enablers	150	grid, cable and electrification suppliers, the bottleneck regardless of who wins	volume growth, not re-rating. Deliberately small: this is the most crowded trade
Structural short / underweight	-150	$P(A) < 0.2$ and $\Delta < 0.05$ bad today and not moving	funds the book; isolates transition alpha from sector beta
Dry powder	250	deploy on a 15% drawdown in the trajectory sleeve within 90 days, else revert to benchmark	indiscriminate post-announcement selling

Three disciplines that follow from the framework and not from taste. *Size by P, not by conviction:* a company at $\Delta = +0.61$ gets more than one at +0.22 because the measure is a probability. *Cap correlated clusters:* names whose transition depends on the same input, cheap hydrogen, a high carbon price, count once against a cluster limit, not three times. *Exit into eligibility:* Pástor, Stambaugh & Taylor (2021) show green assets earn **lower** expected returns in equilibrium, so this is a repricing trade with an endpoint. Sell when P(A) exceeds 0.8.

8.4 What the framework cannot decide, and we will not pretend it can

Our universe is 18 energy companies scored against each other. **It cannot rank Apple against ExxonMobil**, so a real \$1bn book needs the same machinery run per sector with sector-neutral position limits. It has no view on the carbon price, on green hydrogen economics, or on whether transition capex earns its cost of capital, and that last one is decisive: converting high-ROCE fossil assets into regulated-return renewables can be ESG-accretive and value-dilutive at once. **Naming specific non-energy securities would be an opinion dressed as an output.** The rule above is the output; the names in Exhibit 12 are the ones our data actually supports.

Appendix A • The formula

Indices: company $i = 1 \dots 18$, field $f = 1 \dots 138$, year $y = 1 \dots 5$, base slot $b \in \{1,2,3\}$, pillar $P \in \{E, S, G, F\}$. Steps 1 to 5 produce a grade tensor and do not involve the pillar weights.

Step	Operation	Detail
1 · exposure	$x = \text{VAR}/\text{BASE}$ if type Q, else select variant b	one operator per type, so the divisor is applied exactly once
2 · window	latest reported · fixed FY2024 · 3-year mean	reductions run over years the company reported, not calendar columns
3 · direction	Question A level · Question B change	separate grids in v2
4 · orientation	$u = \text{pol} \cdot q$, higher = worse	$\text{pol} \in \{+1, 0, -1\}$; 0 = deliberately neutral
5 · normalise	percentile · winsorised · log min-max · elasticity	elasticity falls back to log below 8 peers
6 · aggregate	$M(x,w,p,s) = (\sum w_j (x_j + s)^p)^{1/p} - s$, $p = \frac{1}{2}$, $s = 10$	$p=1$ recovers MSCI's mean; $p \rightarrow 0$ the geometric
7 · rank	$R = 100(N - \rho)/(N - 1)$	average rank descending; 100 = best
8 · distribute	consensus = median R; contestation = $\text{IQR} \div 5.88$	reported in rank places

Order matters. Aggregate fields *inside* each specification, then take the distribution *across* specifications. Reversing them averages 138 fields first and collapses the spread.

Edge cases the referee asked about, now defined. Neutral polarity sets the field to a constant 50 rather than entering min-max (no division by zero). Log models shift the series positive before taking logs. Change from a zero base returns NaN and the field drops rather than dividing by zero. The elasticity model regresses $\log(\text{impact})$ on $\log(\text{activity})$; §4.2 of v1 stated it with $\log$ intensity on the right-hand side, which was a typesetting error, Appendix A.5 was correct and is what runs.

Appendix B • Referee objections answered in code

Ref	Objection	Verified answer
B1	Contestation with direction and peer set held fixed	3.0 places (Question A × BECS), 5.0 (Question A × sector). Contestation survives.
B2	Are the distributions two-humped?	Yes. EOG: level median 94.1, change 35.3 to 10.0 places apart. Confirmed and fixed by splitting the grid.
B3	What happens to non-producers under the production denominator?	They keep S, G and F and lose only E. All 18 still receive a composite in every production cell. 92 Q-field grades for producers, 0 for non-producers.
B4	How many of the 17,496 cells are distinct?	648 grade-stage cells → 606 distinct , 42 exact duplicates. Under change, a fixed FY2025 base does cancel, as the referee says.
B5	Is contestation group size or evidence?	Partly size. Mean IQR: 2-member 6.0, 3-member 7.7, 7-member 7.0 not monotone. Coverage rank-corr −0.38.
B6	Is direction just the F pillar toggling?	No. Zeroing the F weight moves direction from 18.2% to 15.0% .
B7	Could any company ever be rated robust?	Under the pooled grid, no, which is why the verdict column was uninformative and is removed. Within Question A × BECS, 4 of 18 fall below 2 places.
C2	Which settled statistic do you stand behind?	>95% of Question-A cells: 33 of 153. The 'every method' wording is withdrawn.
C4	Rank spread across the weight presets	Median 3.0 places, max 6.0 (WMB). Reported in Exhibit 10.
D5	Which field does no company report?	DX831 , a governance field. 139 in the registry, 138 with data; the model now uses 138.

Appendix C • Field registry

Pillar	Boolean	Percentage	Quantity	Rate	Structural	Total
Environment	18	4	15	0	0	37
Social	9	0	0	5	0	14

Pillar	Boolean	Percentage	Quantity	Rate	Structural	Total
Governance	25	0	0	0	53	78
Financial	0	0	0	9	0	9
Total	52	4	15	14	53	138

Field weights within a pillar are equal v1 never stated this and the referee is right that it matters: with 9 financial and 78 governance fields, one financial field carries 8.7× the weight of one governance field *within its pillar's contribution*. Pillar weights then renormalise, which is why the pillar weighting in Exhibit 10 moves ranks as much as it does. **Polarity for the 63 uncontested governance fields** follows governance orthodoxy: independence, refreshment, disclosure and audit tenure limits score positively; concentration, entrenchment and long tenure score negatively. The 15 contested fields are listed in the repository's question set.

Appendix D · Sources

MSCI ESG Ratings Methodology, Executive Summary, November 2020, 17pp. We note the limitation the referee raises: this is the public executive summary, not the full methodology, and v1 overstated it as reading the methodology 'end to end'.

Bloomberg ESG Scores Methodology, issued Sep 2023, updated Dec 2025, 58pp. Read in full.

Berg, Kölbel & Rigobon (2022), *Aggregate Confusion*, Review of Finance 26(6), divergence split **56 / 38 / 6**.

Simonsohn, Simmons & Nelson (2020), *Specification curve analysis*, Nature Human Behaviour 4 a specification must test the research question, be statistically valid, and not duplicate another. v1 breached the first and third; v2 addresses both.

Avramov, Cheng, Lioui & Tarelli (2022), *Sustainable investing with ESG rating uncertainty*, JFE, cross-provider dispersion.

Christensen, Serafeim & Sikochi (2022), *Why is corporate virtue in the eye of the beholder?*, The Accounting Review 97(1), more disclosure, more disagreement.

Dimson, Marsh & Staunton (2020), *Divergent ESG ratings*, JPM 47(1), average correlation 0.45.

Pástor, Stambaugh & Taylor (2021), *Sustainable investing in equilibrium*, JFE, green assets earn lower expected returns in equilibrium.

Merton (1987), *A simple model of capital market equilibrium with incomplete information*, Journal of Finance, investor recognition.

Appendix E · Reproducibility

Command	Produces
python3 registry.py	the field registry, printed for inspection
python3 load.py	the arrays and a coverage report
python3 run.py	every specification → out/ (18 seconds)
python3 export_web.py	the pillar cube for the browser explorer
python3 -m http.server 4173 --directory ../web	the interactive tool

numpy and pandas only. The browser explorer reproduces run.py to 0.05 rank points on all 18 companies. **Bloomberg Terminal cell values are withheld** we do not share these indicators and values because they require the proper licensing, and we are happy to show the files in person to demonstrate that the model runs on real data and would scale. Everything derived from them is published: the loader, the formula, all computed outputs, the explorer. The referee is right that this makes full reproduction contingent on a Terminal; the outputs are committed so every figure in this paper can be checked without one.

Appendix F · Roadmap, in the order we would do it

#	Work	Why it is first
1	Place Bloomberg's own ESG score inside each company's distribution	The paper's central claim is that a published score is one draw from a distribution. We had the Terminal and did not test it. Highest value, lowest cost.
2	Rerun on revenue and on a rolling denominator	Removes the weakest defensible choice in the paper.

#	Work	Why it is first
3	Effective-specification weighting	606 distinct of 648; weight cells by distinctness so duplicates stop inflating the distribution.
4	Source Scope 3 and transition capex	The two most material omissions for oil and gas.
5	Cross-sector run with sector-neutral normalisation	Delivers the S&P 500 promise in the title.
6	Backtest the \$8 signal	Does $\Delta = P(B) - P(A)$ predict forward returns or rating migration? Until tested, \$8 is a hypothesis with a computed signal, not a validated strategy.

Disclosures. Prepared for ETHack 2026, Challenge #1, by a student team over 24 hours and revised after referee review. This is not investment research, not a recommendation to buy or sell any security, and was not prepared by a regulated entity. Section 8 is a methodological demonstration on a hypothetical mandate; company names appear as data points. All computed figures derive from the committed pipeline and can be regenerated from source. Characterisations of MSCI's and Bloomberg's methods are ours and any error in them is ours.